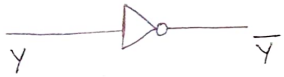
Logic Gates

1) NOT Gate :-



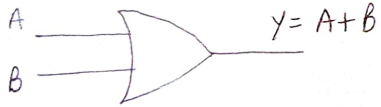
Input Y	Output $\overline{Y}$
0	1
1	0

2) AND Gate :-



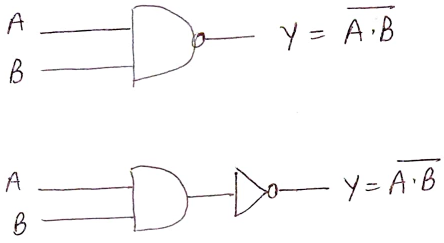
A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1

3) OR Gate :-



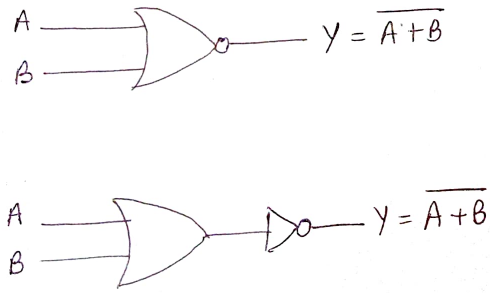
A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1

4) NAND Gate :-



A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0

5) NOR Gate :-



A	B	Y
0	0	1
0	1	0
1	0	0
1	1	0

6) Exclusive - OR Gate (EX-OR) :-

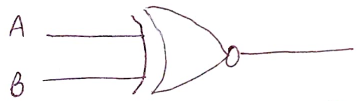


$Y = A \oplus B$

or $Y = \bar{A}B + A\bar{B}$

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0

7) Exclusive - NOR Gate :- (EX-NOR) :-



$Y = A \odot B$

or $Y = \overline{A \oplus B}$

or $Y = \overline{\bar{A}B + A\bar{B}}$

A	B	Y
0	0	1
0	1	0
1	0	0
1	1	1